Corridor, Not Factory: Trade Reorientation and the Missing Investment Response in Kazakhstan, 2022–2025

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Abstract

A large, sector-specific rise in demand for a tradeable good is expected to induce domestic investment. After February 2022, with Russia's direct trade with its main partners disrupted, part of it was reallocated through neighbouring economies, Kazakhstan among them. Using product-level trade data (HS6, 2018–2025) we measure the demand shock, estimate what Kazakhstan retains, and ask whether it induced investment. In 29 product lines, exports to Russia rise discontinuously in May 2022, roughly tenfold; an externally compiled product list not selected on Kazakh outcomes shows the same break. Propagating a wholesale-and-freight margin—6–14%, a band we choose rather than estimate—through the input–output table implies domestic value added of 5–11 cents per rerouted dollar, against about three-quarters for a produced dollar. Across three commercial deal databases we find no investment response in the reoriented lines, though four post-2022 years rule out only a large one. We read the null through three gates: market access, irreversibility, and capital-market institutions. The case identifies an open market-access gate—a customs union makes trade a close substitute for production—and rules out capital-market institutions as

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the binding constraint; whether irreversibility independently binds cannot be settled here. Kazakhstan operates as a corridor, not a factory.

Keywords: trade reorientation, demand shocks, investment under uncertainty, irreversibility, industrial policy, value added, institutional voids, transit economies, Kazakhstan. **JEL:** F14, F15, E22, D25, O14, O24, O33, P33.

1 Introduction

When demand for a country's tradeable output rises sharply and persistently, the textbook expectation is that supply follows: firms expand, new entrants appear, and—if the country was importing the good—some of the demand is met domestically. This logic underlies a large part of development economics, from infant-industry and import-substitution arguments to modern work on global-value-chain entry and the investment effects of trade shocks. Its premise is that a demand signal, once clear enough and durable enough, is acted on by investment.

After February 2022 Kazakhstan received an unusually clean version of such a signal. The disruption of direct trade between Russia and its principal partners reallocated a set of trade flows through third countries. Chupilkin et al. [2026] document the pattern: a sharp fall in EU exports to Russia alongside a matching rise in EU exports to Armenia, Kazakhstan and the Kyrgyz Republic, with product-level and mirror-statistics evidence of a corresponding rise in those countries' exports to Russia; Chupilkin et al. [2025] quantify the substitution. A policy literature documents the same reallocation [Simola, 2023, Bilousova et al., 2024, Wiśniewska, 2025]. For Kazakhstan the reorientation is concentrated in a well-defined set of technology-intensive products and is large: at the monthly level, exports to Russia in these lines rose roughly tenfold. The routes, the demand and the products are all observable in the trade data. We treat this as a case study—a large, sector-specific, well-identified demand shock—and ask whether it induced the domestic investment response the textbook predicts.

This paper asks what Kazakhstan did with that signal. We proceed in three steps.

First, we measure the shock. Using UN Comtrade at HS6, annual for 2018–2025 and monthly for 2019–2025, we identify from the data a "surge basket" of 29 product lines in which the outflow to Russia and the inbound flow (Western reporters plus China) both at least doubled from 2019–2021 to 2022–2024. The outbound aggregate breaks sharply in 2022m5 (monthly sup-F of 561); the inbound flow breaks in the same month but more weakly, and not at all in the noisier annual series, where a large pre-existing China flow dominates. Armenia and the Kyrgyz Republic show the same 2022 break with no comparable domestic reform. A difference-in-differences against control lines gives $\gamma = 2.44$ on the data-driven basket—but a selection-rule-matched permutation places the estimate inside its own null distribution, so we do not read that coefficient as an identified magnitude. The externally compiled priority list, which is not selected on Kazakh outcomes, gives the same sign and pattern; a placebo on the largest civilian lines runs the other way.

Second, we measure what Kazakhstan retains. On the incremental flow to Russia (about \$479m over 2022–2025), the unit-value and weight evidence does not sharply distinguish the re-export flow from ordinary Kazakh trade. Propagating a wholesale-and-freight margin—6–14%, a band we choose rather than estimate—through the input–output table (OECD ICIO) implies domestic value added of about **5–11 cents** per dollar rerouted, against roughly three-quarters for a dollar of domestic manufacturing output. Because the input–output step does almost no work, the headline scales one-for-one with the margin: it stays below one-fifth of a produced dollar for any margin below a fifth (Section 6). Customs duty on the incremental flow is well under 1% of customs revenue, and the flow is 0.2–0.3% of GDP.

Third, we ask whether the shock induced investment—and find no response the data can detect. We assemble deal-level data on mergers, acquisitions, and private-equity and venture transactions in Kazakhstan for 2015–2025 from three commercial databases (Capital IQ, PitchBook, Preqin), together with the aggregate investment activity of Kazakhstan's state investment corporation as reported in Qazaqstan Investment Corporation et al. [2026]. Deal-making in manufacturing of tradeable goods, transport and logistics, and distribution runs at essentially its pre-2022 rate—if anything lower after 2022; with four post-2022 years the test rules out a large response, not a modest step-up, and the null is in any case over-determined—the January 2022 unrest, the cross-border compliance and reputational exposure that transactions of this kind carry, and the nationalisation wave would each depress entry. Reported deal value rises, but is concentrated in ownership transfers of existing or distressed assets—two of the largest are nationalisations—while greenfield and new-capacity transactions show no post-2022 increase.

We interpret the non-response with a framework in which the supply response is the product of three gates: *irreversibility* (does anyone build for a shock this uncertain and policy-contingent, given the option to serve the demand through trade?), *market access* (is domestic transformation required to reach the final market?), and *institutions* (can the projects that clear the first two be financed?). For Kazakhstan the market-access gate is plainly open—a customs union with the destination makes trade a close substitute for production—which is close to sufficient for a null, and which also forgoes the technology transfer a rules-of-origin regime would have forced [de Souza et al., 2024]. The case also rules out capital-market institutions as the binding constraint for this particular null: the state investment corporation, which faces neither an external-finance nor an exit constraint, did not build for the reallocation either. Whether the irreversibility gate *independently* binds we cannot settle here—a within-country comparison (an assembly response to the durable passenger-vehicle demand shock, none to the re-export shock) is suggestive but confounded, and the re-export flow's persistence through 2025 makes it look uncertain rather than clearly transitory. The institutional gate does explain a separate fact—why the durable adjacent opportunities the shock reveals go unfinanced by private capital.

Contribution. The paper makes three contributions. (i) Its primary contribution is empirical: to our knowledge it is the first host-economy analysis of what a transit economy retains from a post-2022 trade reallocation and of whether it induced investment, extending the existing literature—which establishes that the reallocation happened and its scale—to what the intermediary retains, and to 2025. It shows that a large, sector-specific tradeable-demand shock produced no detectable investment response, a fact that speaks to when trade shocks induce a domestic supply response and when they do not [Juhász, 2018, Bloom et al., 2016, Juhász et al., 2024, Gopinath and Neiman, 2014]. (ii) It organises the two standard readings of a missing supply response—investment under uncertainty and irreversibility [Dixit and Pindyck, 1994], and institutional voids in the capital market [Khanna and Palepu, 1997, 2000, Rajan and Zingales, 1998]—into a single decomposition, and adds a customs-union market-access gate under which trade substitutes for production. The case cleanly establishes an open market-access gate and rules out the institutional gate as the binding constraint for this null; whether irreversibility independently binds it cannot determine. The decomposition is an organising device, not an estimated model. (iii) Methodologically, it combines public trade data, an input–output table, and multi-source commercial deal data into a replicable measure of what a transit economy captures, cross-checking the deal counts across databases rather than relying on one.

The rest of the paper is organised as follows. Section 2 describes the setting; Section 3 sets out the framework; Section 4 the data, including the deal-database construction; Section 5 measures the reorientation; Section 6 what Kazakhstan retains; Section 7 the (non-)investment response; Section 8 asks which gate binds; Section 9 the moderators and external validity; Section 10 where investment would pay off; Section 11 draws the policy implications; and Section 12 concludes.

2 Setting

Kazakhstan is a founding member of the Eurasian Economic Union and, before it, the 2010 customs union with Russia and Belarus, which removes the customs frontier between Kazakhstan and Russia: goods that clear customs on entry to Kazakhstan move onward to Russia without a further border, and a supply from Kazakhstan to Russia is treated as intra-union. The union aligned Kazakhstan's external tariff schedule with Russia's and measurably reshaped the level and composition of its imports [Isakova et al., 2016]. Kazakhstan is also the land bridge between China and Russia and, via the Trans-Caspian corridor, between China and Europe, though as a doubly landlocked economy its overland logistics costs are high and its supply-chain reliability limited [Arvis et al., 2010]. These features make it a natural transit route for goods re-exported without transformation: because it shares a customs union with the destination, serving the final market from Kazakhstan requires no domestic processing.

The land route has also become more valuable for reasons unconnected to any single trading partner. Since 2022 the northern land corridor through Russia has been closed to most Western supply chains, and over 2023–2025 the principal maritime routes were disrupted in turn—draft restrictions on the Panama Canal, re-routing around the Red Sea, and periodic risk in the Strait of Hormuz. Each raises the value of an overland China–Europe alternative, and the shortest route that avoids both Russia and the maritime chokepoints runs through Kazakhstan and across the Caspian. We return to what this implies for investment policy in Section 11.

Several things other than the reorientation changed in Kazakhstan around 2022, and each is a potential confound. The first is the "New Kazakhstan" reform programme associated with the presidency of Kassym-Jomart Tokayev, an outward-oriented agenda that gathered pace in 2022 with a June constitutional referendum and a November snap election. A reform of that kind would raise trade and investment broadly and gradually. What we observe is the opposite: a response concentrated in a narrow set of products, discontinuous in mid-2022, with a civilian control basket that moves far less and the same 2022 break in Armenia and the Kyrgyz Republic under no comparable reform.

Three further shocks bear specifically on the *investment* null and cannot be differenced away: the January 2022 unrest and state of emergency, in the first month of the post-period; the cross-border compliance and reputational exposure that transactions of this kind carry, which deters exactly the foreign acquirers who might have answered the shock; and the 2022–2025 wave of nationalisations, which both crowds the state investment pipeline and signals ownership risk to private entrants. We return to these where they matter, in Sections 7 and 8. They make the investment null *over*-determined rather than fragile, but they also mean we cannot cleanly attribute it to the trade shock alone.

3 A framework: when a demand shock builds capacity

Consider a potential entrant deciding whether to build capacity to serve a newly enlarged demand for a tradeable good that the country currently imports. The good can be re-exported to the final market without domestic transformation—this is the relevant case for a member of a customs union with the destination. The entrant has two ways to serve the demand. It can *trade*: buy the good abroad and resell it into the final market, earning a margin μ_T per unit, with no sunk cost and the ability to stop costlessly if demand falls. Or it can *produce*: sink an irreversible cost I to build capacity, then earn a margin μ_P per unit on domestic output for as long as it operates, with the capacity worth only scrap if the demand disappears.

Let the enlarged demand support quantity Q per period and persist each period with probability ρ (so expected duration $1/(1-\rho)$), and let r be the discount rate. Trading is always weakly profitable while the demand lasts, with present value $V_T = \dfrac{\mu_T Q}{1-\rho/(1+r)}$. Building is profitable only if

$$\underbrace{\frac{\mu_P Q}{1-\rho/(1+r)}}_{\text{PV of production margin}} - I > \underbrace{\Omega(\sigma,\rho)}_{\text{option value of waiting}} + V_T, \tag{1}$$

where Ω is the value of the option to defer, increasing in the uncertainty σ around the shock's size and duration and decreasing in ρ [Dixit and Pindyck, 1994]. Four observations follow.

(i) The irreversibility gate. The right-hand side of (1) is large exactly when the shock is transitory (low ρ), uncertain (high σ, e.g. because it is policy-contingent), and the technology is sunk-cost-intensive (high I). When any of these is extreme, no entrant builds—$R \approx 0$—regardless of the other conditions. This is a statement about *whether anyone builds*, and it does not depend on who the entrant is or how deep the capital market is.

(ii) Trade is the outside option, not just "wait". Because the good is re-exportable, the alternative to building is not idleness but V_T: the entrant can serve the demand through trade indefinitely. The demand shock raises the return to *both* margins. When the shock is transitory and μ_T is available at low cost, (1) fails and the supply response is trade rather than production. A corollary: if $\mu_P \leq \mu_T$—if Kazakhstan cannot even charge more for a transformed good than it pays for the input—then building is dominated before Ω is added. We do not identify μ_T from the unit-value evidence (Section 6); what indicates $\mu_P \lesssim \mu_T$ is instead the sector's near-zero domestic base (≈\$230m of electronics output) and its low value-added multiplier (0.69; Section 10), which put a domestic plant's transformation margin below the thin markup a re-exporter already earns.

(iii) The market-access gate. If serving the final market instead *required* domestic transformation—binding rules of origin, local-content rules, or simply a customs border that a pure re-export cannot cross—then $V_T \to 0$ and (1) is far more likely to hold. Membership of a customs union with the destination opens this gate and is what makes trade a viable substitute for production. The same logic runs through the imported-input intensity of the would-be domestic producer: where domestic production would itself lean heavily on imported components—the roundabout structure of Gopinath and Neiman [2014]—the value a local plant adds over a pure re-export is smaller still, and the transformation margin $\mu_P - \mu_T$

narrower. The gate also governs *dynamic* gains. de Souza et al. [2024] show, using Brazilian firm-to-firm technology-transfer records, that a foreign supplier facing a lower tariff shifts from *transferring* technology into the market toward *exporting* goods to it, and that learning from imported goods is less efficient than learning from a transfer—so import liberalisation raises static welfare but slows diffusion and productivity growth. An open market-access gate is the limiting case: with a pure re-export, the foreign firm has no reason to transfer anything, and the host forgoes not only the static value added of Section 6 but the learning that would come with a plant.

(iv) The institutional gate. Suppose (1) *does* hold for some project—a durable opportunity with modest sunk costs. It is still realised only if the entrant can finance I and, for an outside equity investor, exit. Where the capital market lacks growth equity and a functioning exit, projects that clear the irreversibility gate are not funded; the capital that does move is captive—retained earnings, or a state or development-finance investor—and is allocated by criteria other than R [Khanna and Palepu, 1997, 2000, Rajan and Zingales, 1998].

The three conditions can be summarised as a product. Each observation above identifies a factor that, on its own, is enough to drive the investment response to zero: an extreme irreversibility configuration, an open market-access gate (transformation not required, so V_T is large), or a capital market that cannot fund the projects that pass. Writing

$$R = \underbrace{f\big(\rho,\ \sigma^{-1},\ I^{-1}\big)}_{\text{irreversibility}} \times \underbrace{h(\text{transformation required})}_{\text{market access}} \\ \times\ \underbrace{g(\text{financial depth, exit, private/state mix})}_{\text{institutions}} \tag{2}$$

makes this explicit: $R \approx 0$ whenever any factor is near zero. This is an organising device, not a result derived from (1)—the threshold there is on a sum, and the factors interact inside an indicator—and its content is that the three conditions are separately necessary. The first two bear on whether production is ever the right way to serve the demand, the third on whether the projects that pass get built. Sections 5–8 show that for Kazakhstan the market-access gate is open—close to sufficient for $R \approx 0$ on its own—and rule out the capital-market institutional gate as the binding constraint for this null; whether the irreversibility gate independently binds the case cannot settle. Section 9 and Section 10 show that the institutional gate does explain a separate fact—the durable adjacent opportunities the shock reveals go unfinanced.

A mediation restatement. The same structure can be put in mediation terms; we do so as an interpretive device, not as a model we estimate—the data below have no firm-level panel linking the trade response to the investment response. Let the demand shock X bear on the investment response Y through a direct path $X \to Y$ and an indirect path $X \to M \to Y$, where the mediator M is the volume of re-export activity the shock generates. The shock raises M; higher M substitutes for domestic capacity, so it works against Y. When the market-access gate is open ($h = 1$, no transformation required), trade is a close substitute and nearly all of the shock is carried by M: the channel that would raise Y is barely opened, and the net effect on investment is close to nil. When the gate is closed ($h < 1$), M absorbs less of the shock and part of X bears on Y directly, with that residual governed by the irreversibility parameters (ρ, σ, I)—whether the un-absorbed demand is large and durable enough to clear (1). Kazakhstan is the polar case: $h = 1$, and the unit-value wedge of Section 6 is a rough estimate of the margin on the traded path, which is no higher than the cost of the input—so little is left for the direct path even before the irreversibility gate is applied.

4 Data

Trade. UN Comtrade, HS 6-digit. Annual 2018–2025 for magnitudes; monthly 2019m1–2025m12 (authenticated API) for break timing and the event study. Kazakhstan stopped reporting monthly to Comtrade after February 2024, so the monthly series uses 2019–2023 and 2025; 2024 is covered by the annual data and by mirror flows. We use Kazakhstan's reported exports to Russia (complete) and, for the inbound side, a partner-reported (mirror) measure: the sum of exports to Kazakhstan reported by the EU-27, the United Kingdom, the United States, Japan, Korea, Switzerland, Norway ("Western") *and China*, because Kazakhstan's own import reports are incomplete for 2020–2022. China supplies about two-thirds of this inbound value and its flow to Kazakhstan was large and rising well before 2022; we therefore also track the Western component on its own, since that is the part that moves with the reorientation. Mirror data overstate Kazakh absorption for goods declared for Kazakhstan and then moved onward—the onward-movement bias that Chupilkin et al. [2026] use to identify the reorientation in the first place—a bias we return to in Sections 5.3 and 6. Neighbour comparisons (Armenia, the Kyrgyz Republic) and the economy-wide imports-by-chapter series used in Section 10 are from the same source.

Product sets. We take 75 HS6 lines: 50 codes from an externally compiled list of technology-intensive HS6 items published in February 2024 (the "priority list") plus 25 clearly civilian control codes. We define the **surge basket** from the data as the 29 lines in which the mean inbound flow (West + China) and the mean outflow to Russia each at least doubled from

2019–2021 to 2022–2024, and exceeded \$0.2m (inbound) and \$0.1m (outbound) in the post period, computing the ratio with \$10,000 added to numerator and denominator so that a line rising from near-zero does not qualify on a rounding artefact. Twenty-four of the 29 lines are on the priority list; the codes, tiers and control set are in Appendix B. Because the basket is selected on the post-period outcome, we also run every specification on the priority list, which is defined on product characteristics rather than on Kazakh flows—though, compiled in 2024, it is not independent of the phenomenon it helps describe—and we benchmark the difference-in-differences against a selection-rule-matched permutation test (Section 5.3).

Input–output. OECD Inter-Country Input–Output tables, 2023 edition, Kazakhstan block, 45 ISIC rev. 4 industries, 2019. We form the domestic technical-coefficient matrix A_d from the Kazakhstan-on-Kazakhstan intermediate block, the domestic Leontief inverse $L_d = (I - A_d)^{-1}$, and the direct domestic value-added coefficients v_d; the domestic value-added multiplier for sector j is $(v_d' L_d)_j$. As a robustness check we repeat the calculation on the Kazakhstan Bureau of National Statistics symmetric input–output table (up to 68 products, 2023 reference year).

Deal data. Deal-level records of mergers, acquisitions, private-equity buyouts and growth transactions, and venture rounds involving Kazakhstan-domiciled targets, 2015–2025. The extract combines Capital IQ, PitchBook and Preqin ($n \approx 500$ after de-duplication), each record carrying its native source deal identifier, classified to sectors and flagged for greenfield versus ownership transfer. The exact query specification for the three databases (geography filter, deal types, date range, industry taxonomy, extraction date) is in Appendix A, and Table 3 reports each one's coverage. All are commercial but widely available under academic licence, and the replication package lists the deal identifiers, so the extract is reconstructible. QIC investment activity is taken from Qazaqstan Investment Corporation et al. [2026], which reports it in aggregate (about 98 investments, roughly \$2.2bn, 2015–2025), by sector and year, and names individual projects in its appendix and case studies; no project-level QIC register is public, and the paper uses only the report's aggregate, sector-timing and named-project evidence.

Macro. World Bank World Development Indicators.

Table 1: Surge-basket trade, USD million per year. Inbound is a partner-reported (mirror) measure. China's 2024 figure is large and preliminary; its 2025 figure is not yet reported, so the 2025 West + China total is the Western component only.

	2018	2019	2020	2021	**2022**	2023	2024	2025
West + China → Kazakhstan (mirror)	394	427	483	470	**837**	1,373	2,360	363
of which Western	162	162	168	184	**337**	444	419	363
Kazakhstan → Russia	17	12	7	8	**128**	145	119	133

5 The demand shock

5.1 Magnitudes

Table 1 reports the surge basket. Kazakhstan's exports to Russia in these lines were \$7–17m per year in 2018–2021; they were \$128m in 2022 and \$119–145m in 2023–2025—a rise of roughly an order of magnitude, and the sharpest, cleanest part of the picture. The inbound flow (West + China) ran near \$440m per year before 2022 and roughly \$0.8–2.4bn after, but that series is dominated by China, whose exports to Kazakhstan were already large and rising before the war (and are only partly reported for 2024–2025). The *Western* component of the inbound flow—the part aligned with the reorientation—rose from about \$170m per year to \$340–440m, a factor of about 2.3.

5.2 Timing and the event study

Monthly, the outbound series (Kazakhstan's exports to Russia) breaks sharply in 2022m5 (Bai–Perron, 95% CI 2022m4–2022m6; sup-F of 561, homoskedastic covariance, against critical values in the low tens), following the March-2022 sanctions with a short lag and preceding every political-calendar date. The inbound series (West + China) breaks in the same month (95% CI 2022m4–2022m6), with a second, smaller break in late 2023 as trade-monitoring measures tightened; the Western component alone breaks in 2022m5 more sharply. On the *annual* panel the outbound and Western-inbound breaks are again dated to 2022, while the West + China inbound break is not significant, because the large pre-existing China flow swamps the 2022 movement at annual frequency—one more reason the reorientation is best seen in the outbound series and the Western inbound. The sup-F is computed without a HAC correction and the series is highly persistent, so we read it as evidence of a break and its *date*, not as calibrated inference on the break's size; the event study and the difference-in-differences carry the quantitative weight. Figure 1 plots the monthly series; Figure 2 is a difference-in-differences event study at the HS6×year level (surge basket vs. control lines,

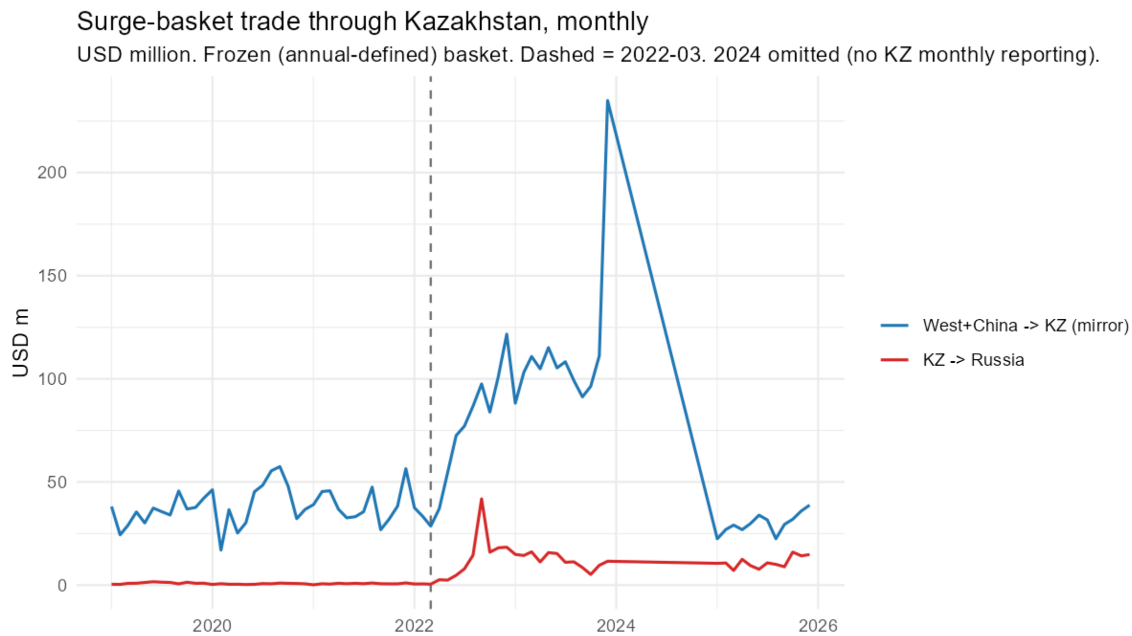

Figure 1: Surge-basket trade through Kazakhstan, monthly. USD million; blue = West + China mirrored exports to Kazakhstan, red = Kazakhstan's exports to Russia. 2024 omitted (no Kazakhstan monthly reporting to Comtrade). Dashed line: March 2022. Source: UN Comtrade.

reference year 2021, 95% confidence intervals clustered by HS6). The pre-treatment coefficients are close to zero and jointly insignificant ($p = 0.78$ outbound, $p = 0.42$ inbound); the outbound coefficient jumps in 2022 and stays elevated through 2025 at about +2.4 asinh points (the Poisson level estimate is larger and is reported in Table 2; we do not read it as a headline magnitude, for the reasons in Section 5.3); the inbound coefficient follows a similar path with wider intervals. That the break precedes the June referendum, that the response is confined to a narrow product set, and that Armenia and the Kyrgyz Republic show the same 2022 break with no comparable reform (Section 5.3) are what separate the reorientation from the "New Kazakhstan" agenda.

5.3 Difference-in-differences

Specification and identifying assumptions. On the annual panel we estimate $\text{asinh}(y_{it}) = \gamma\,(\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$ with HS6 and year fixed effects and standard errors clustered by HS6 (75 lines, 8 years; reference year 2021). Zeros are common on the outbound leg before the shock and rare after—23% of surge-basket cells are zero in 2018–21 against 3% in 2022–25—so we report the asinh specification alongside a Poisson (PPML) estimate in levels, which handles the zeros without a transformation and weights by size; the two tell the same story (Table 2). Identification requires parallel paths absent the shock (a joint test

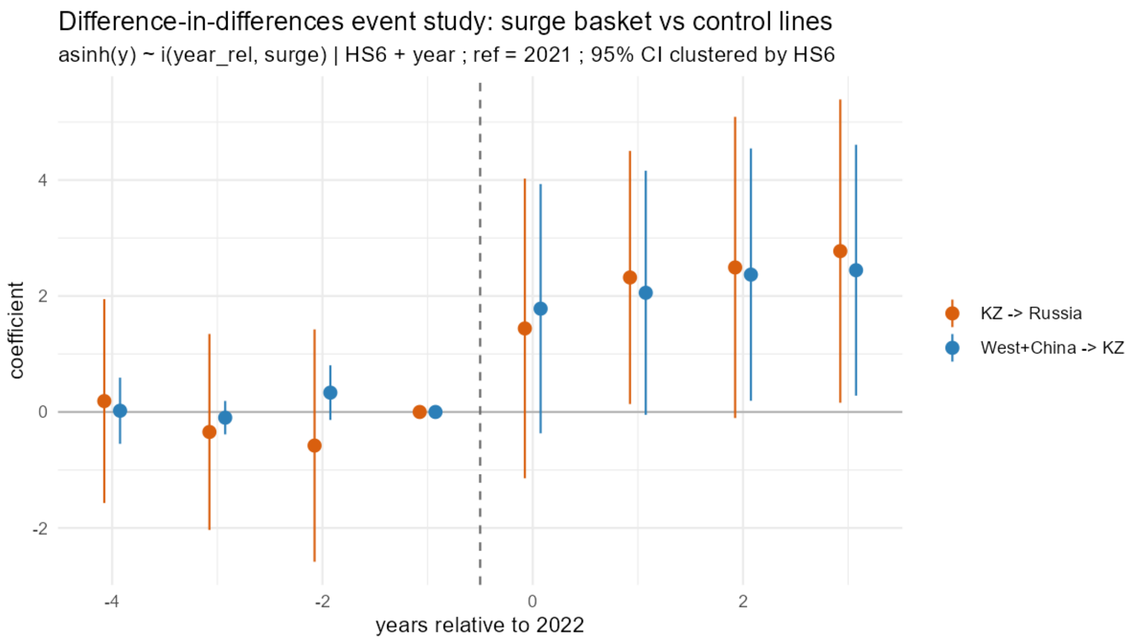

Figure 2: Difference-in-differences event study, surge basket vs. control lines: $\text{asinh}(y_{it}) = \sum_{k\neq-1} \beta_k\,(\text{surge}_i \times \mathbf{1}[t-2022=k]) + \mu_i + \lambda_t + \varepsilon_{it}$, HS6 and year fixed effects, 95% confidence intervals clustered by HS6, reference year 2021. Source: UN Comtrade.

of the three pre-period interaction coefficients does not reject: $p = 0.42$ inbound, $p = 0.78$ outbound) and no anticipation (dropping 2022 entirely, or setting the treatment year to 2023, leaves γ essentially unchanged; see below). The civilian control lines are not perfectly clean—their exports to Russia also break in 2022 (Section 5.3), so γ is if anything attenuated relative to the economy-wide effect. The surge basket is *selected on the post-period outcome*, which inflates γ mechanically; the selection-rule-matched permutation test below, not the cross-sectional regression, is what speaks to this and to the cross-sectional dependence a single common treatment date induces. We also report wild-cluster-bootstrap p-values (few clusters, within-HS6 serial correlation) and Holm-adjust across the four surge-basket outcomes. Both the analytic and the bootstrap standard errors cluster within HS6 but treat lines as independent; with a single common treatment date and a shock that hits related product lines together, that is optimistic. The randomisation-inference p-values below do not rely on it—they resample whole treatment histories—and are the inference we lean on.

Results. Table 2 reports γ for the surge basket, the externally defined priority list, and a placebo. On the surge basket, exports to Russia rise by $\gamma = 2.44$ ($p = 0.013$; wild-bootstrap $p = 0.010$; Holm-adjusted $p = 0.039$; Poisson in levels 3.5×, $p = 0.02$). The inbound flow (West + China) rises by $\gamma = 2.10$ ($p = 0.05$, not significant after Holm adjustment); the Western component alone by 1.73 ($p = 0.08$); Kazakhstan's own reported imports by 1.88

($p = 0.004$, Holm $p = 0.016$). The externally defined priority list, which is not selected on the Kazakh outcome, gives the same pattern ($\gamma = 2.88$ outbound, 2.12 inbound). The outbound effect survives two demanding checks: adding size-decile×year fixed effects (pre-2022 inbound size) leaves $\gamma = 2.29$ ($p = 0.003$), and dropping 2022 raises it to 2.71 ($p = 0.012$). The same size×year controls cut the West + China inbound coefficient to 1.28 ($p = 0.05$), consistent with the inbound leg being the weaker of the two throughout. Tightening the selection threshold from a doubling to 2.5× or 3× raises the outbound γ to about 3.1; loosening it to 1.5× (40 lines) lowers it to 1.6 ($p = 0.08$). A leave-one-HS2-out jackknife shows the effect is carried by HS 85: dropping it (13 of the 29 lines) leaves $\gamma = 1.36$ ($p = 0.21$); the other six chapters, dropped one at a time, give $\gamma = 2.51$ (drop HS 39), 2.48 (HS 40), 2.67 (HS 61), 2.28 ($p = 0.06$; HS 84), 3.16 (HS 90) and 2.59 (HS 96), all significant at 5% except HS 84 at 6%. The point estimate thus depends materially on the electrical-machinery lines, which is where the reorientation is concentrated; the sign and the sub-5% significance do not, except when the largest chapter is removed.

Three qualifications keep us from resting identification on this cross-sectional regression (the full battery is in Appendix C). First, the selection-rule-matched permutation: when we impose the null by permuting the year labels within each HS6 and re-applying the doubling rule, the rule selects on average five lines (never as many as the 29 observed, so the *existence* of the surge basket is not a chance artefact, $p < 0.001$); a trend-preserving version that only shifts the break off calendar-2022, keeping each line's autocorrelation and level trend, gives the same result (null mean five lines, $p < 0.001$), so it is the 2022 alignment and not a pre-existing trend that the selection picks up. But the observed $\gamma = 2.44$ sits squarely inside the rule-matched null distribution of the coefficient itself (null mean 2.79, s.d. 1.50; $p = 0.58$ for $|\gamma^{\text{perm}}| \geq |\gamma^{\text{obs}}|$), so the coefficient's *magnitude* is not separable from the selection and we do not read it as an identified effect size. Second, the two selection thresholds bind differently: 55 of the 75 candidate lines clear the outbound doubling but only 41 clear the inbound one, so requiring both is a real screen on the import side, not just on the outflow to Russia—a partial check against selecting the basket purely on the outbound outcome the difference-in-differences then uses. Third, the placebo row: assigning fake treatment to the largest civilian lines gives $\gamma = -0.95$ ($p = 0.001$) – the large consumer-goods lines contracted after 2022 as the tenge depreciation would predict – which shows the control pool is not homogeneous in trend. The surge coefficient's survival of the size-decile×year controls is what addresses this; we do not read the placebo as corroboration. Taken together, the cross-sectional difference-in-differences documents the outbound reorientation but does not identify it on its own; the structural breaks (Section 5) and the neighbour comparison below carry that weight. The mirror-gap outcome (partner-reported inflow minus Kazakh-reported

Table 2: Difference-in-differences: γ on treated$\times$post in $\text{asinh}(y_{it}) = \gamma(\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$, HS6 and year fixed effects, cluster-robust standard errors (by HS6) in parentheses. Surge-basket and priority-list rows have $N = 600$ (75 lines $\times$ 8 years); the placebo row has $N = 200$ (the 25 civilian lines). The surge basket is selected on the inbound (West + China) and outbound flows; the priority list (50 codes) is externally defined.

Treated set	Exports to Russia	Inbound (W+China)	Inbound (Western)	KZ-reported imports
Panel A. asinh DiD, γ (cluster-robust s.e. by HS6)				
Surge basket (29 HS6)	2.44 (0.96)*	2.10 (1.06)	1.73 (0.98)	1.88 (0.63)**
Priority list, external (50 HS6)	2.88 (0.74)***	2.12 (0.71)**	1.93 (0.67)**	2.94 (0.53)***
Placebo (largest civilian lines)	—	−0.95 (0.25)**	—	—
Panel B. Poisson (PPML) DiD in levels, e^{β} (multiplicative)				
Surge basket (29 HS6)	3.5×*	2.4×***	—	—
Panel C. selection-rule-matched randomisation inference (surge basket, exports to Russia)				
rule-matched null γ: mean 2.79, s.d. 1.50; $P(\lvert\gamma^{\text{perm}}\rvert \geq 2.44) = 0.58$				
basket-existence null: $P(\text{rule selects} \geq 29 \text{ lines} \mid H_0) = 0.000$				

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (cluster-robust; before multiple-testing adjustment). After Holm adjustment across the four surge-basket outcomes, exports to Russia and KZ-reported imports remain significant at 5%. Surge-basket and priority-list rows in Panel A have $N = 600$ (75 lines $\times$ 8 years); the placebo row has $N = 200$. Panel B: exact PPML coefficients $\beta = 1.26$ ($p = 0.02$) outbound and $\beta = 0.85$ ($p < 10^{-4}$) inbound. Panel C: the coefficient's *magnitude* is not separable from the post-outcome selection, but the *existence* of a 29-line basket is not a chance artefact under either a free or a trend-preserving permutation.

imports) does not move ($\gamma = 0.54$, $p = 0.81$).

The reform confound. Three things separate the reorientation from the outward-oriented policy agenda of 2022. First, the civilian control basket tells the story a broad reform cannot: on the annual data its inbound flow shows no 2022 movement at all, and while its exports to Russia do shift—a general Russia-trade effect—the shift is far smaller than the surge basket's on the same test, and the monthly surge-basket outbound break is much sharper still; the cross-sectional placebo runs the wrong way. Second, the outbound break is dated 2022m5, before the June referendum and the November election. Third, and most directly: Armenia's surge-basket exports to Russia rose from \$7–15m per year to \$71m in 2022 and \$93m in 2023, and the Kyrgyz Republic's from \$1–7m to \$12m and \$30m, on the same timing—neither country having a "New Kazakhstan" reform. The direction is also asymmetric—inbound from Western reporters, onward to Russia.

6 What Kazakhstan retains

6.1 The retained trade margin

We measure what Kazakhstan retains on the *incremental* flow—exports to Russia above the pre-2022 baseline, about \$479m cumulatively over 2022–2025—since that is the part the reorientation drove. The \$479m increment is measured against a flat counterfactual at the 2018–21 mean (\$11m/yr); because the pre-2022 outbound series is itself declining (\$17m→\$8m), that is the conservative choice—extrapolating the pre-trend would put the counterfactual near zero by 2024 and raise the increment to about \$521m, widening every downstream figure by roughly a tenth. Set against the incremental *Western* inbound of about \$887m, the flow-through is roughly one half; set against the incremental West + China inbound of about \$3.2bn, it is one sixth.[1] The inbound leg is the more weakly identified of the two—its DiD coefficient loses significance under size×year controls and after multiple-testing adjustment (Section 5.3)—so the flow-through ratio should be read as indicative, not precise. The remaining \$887m − \$479m = \$408m of the Western increment decomposes into only about \$41m of onward exports to destinations other than Russia (Kazakhstan's surge-basket exports to the rest of the world barely move) and about \$367m of domestic absorption, inventory and the c.i.f./mirror measurement gap—so the flow-through gap is not hidden re-export through third countries. The Western basis is the relevant one for the reorientation, and we use it here.

The unit-value evidence is consistent with a corridor and does not identify a retained margin on its own. On a c.i.f./f.o.b. basis—Kazakhstan's f.o.b. re-export price against the c.i.f. price it records paying on import, which includes freight into a landlocked economy—the median ratio across matched HS6×period cells is **0.73** (annual, $n = 112$; 0.70 monthly). On a like-for-like f.o.b. basis, against the exporter-reported price at the partner's border, the median is **1.6** (annual; 1.5 monthly), the difference being roughly the inbound freight. Neither ratio is distinctive—the civilian control basket gives 0.59 and 0.94 on the two bases—and the pass-through regression is weak on both (a slope of 0.06–0.48, significant only in the annual f.o.b. specification). Two features cut against domestic transformation, though neither is decisive. First, the weight ratio— re-exported tonnage over imported tonnage in the matched cell—is 0.08 at the median and 0.11 in aggregate, close to the matched-cell value flow-through (median 0.15, aggregate 0.11); the aggregate weight ratio and the aggregate

[1] Three flow-through ratios appear in this section, on different samples and bases: the *incremental* ratio against the Western inbound (≈ 0.54) and against West + China (≈ 0.15, "one sixth") used here, both on aggregate 2022–25 minus baseline; and the *matched-cell* aggregate ratio (≈ 0.11, sum of re-export value over sum of inbound value across the 112 matched HS6×period cells) used in the weight-ratio discussion below.

value flow-through are the same statistic on different units and coincide here at 0.11. A sub-one weight ratio is therefore largely a flow-through fact—only part of the imported tonnage is re-exported to Russia at all—not evidence on weight gain per transiting unit. Eleven per cent of cells do carry a weight ratio above 1.05, on a wide interquartile range (0.02 to 0.49); under the flow-through reading that upper tail is timing and inventory rather than processing. Restricting to the six near-pure-transit HS6×period cells where value flow-through is near one, the weight ratio is 0.74 at the median (0.26 and 0.94 at the quartiles, on $n = 6$)—no sign of local input added on the goods that do transit, but on a thin slice of the basket. The Comtrade extract as pulled carries value and net weight but not physical quantity, so a cleaner per-unit transformation test is not available here. Second, the dispersion in the f.o.b. wedge is not a processing signal: the high multiples—a median near four in the highest priority tiers,[2] on about half the matched gross flow (Figure 3)—fall in exactly the lines where scarcity markups on goods that became harder to source directly are expected. This is the pattern Feenstra and Hanson [2004] document for Hong Kong's entrepôt trade in Chinese goods, where the re-export markup is largest precisely where the intermediary adds matching and scarcity value rather than physical transformation; the wedge here is if anything smaller and noisier. The value-weighted aggregate wedge is below one, dominated by a handful of high-inbound-value cells and by outbound under-invoicing [Fisman and Wei, 2004]. The wedge does not pin the retained margin, which we take from the national accounts next.

6.2 Input–output propagation

Let m be the trade-and-freight margin Kazakhstan retains on a transiting good as a share of its value, $\bar{v}^{TT}$ the mean domestic value-added multiplier for its trade, transport and warehousing sectors, and $\bar{v}^{M}$ that for manufacturing. Domestic value added per dollar of gross rerouted flow is $m\,\bar{v}^{TT}$; per dollar of domestic manufacturing output it is $\bar{v}^{M}$ [Koopman et al., 2014, Johnson and Noguera, 2012]. From the OECD ICIO Kazakhstan block, $\bar{v}^{TT} = 0.79$ and $\bar{v}^{M} = 0.76$—close enough that the propagation step does little work: the result is essentially m against the full domestic value $\bar{v}^{M}$.

We take m to be what a trading intermediary retains when it buys a good wholesale and re-sells it without a retail leg or a physical transformation—the c.i.f./f.o.b. freight and insurance wedge plus a wholesale (not retail) trade margin. The Kazakhstan Bureau of

[2] The priority list is published with the goods grouped into tiers (Tier 1–2 the most sensitive, Tiers 3–4 progressively less so); we carry the published tier labels unchanged and use them only to describe where in the list the high wedges fall. Tier 1 = HS 8542 (integrated circuits); Tier 2 adds transceivers, capacitors, connectors; Tiers 3–4 are broader machinery, instruments and components.

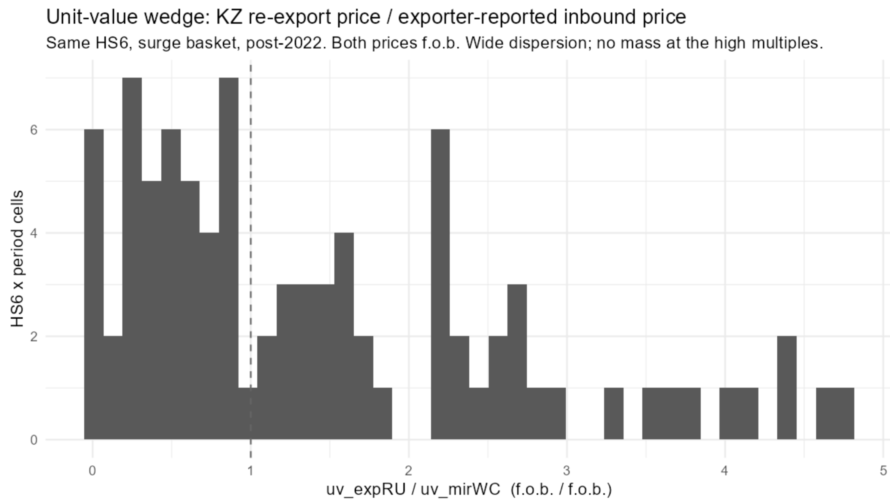

Figure 3: Unit-value wedge: Kazakhstan re-export price over the exporter-reported inbound price (f.o.b. basis), same HS6, surge basket, post-2022. Wide dispersion, with the high multiples concentrated in the most-restricted tiers—the pattern a scarcity markup produces, not the even, across-the-board wedge that a processing margin would.

National Statistics resources table [Bureau of National Statistics, Agency for Strategic Planning and Reforms of the Republic of Kazakhstan, 2024] bounds it: the *transport* margin on machinery and electronics is about 1% of basic-price value, and the *full* trade-and-transport margin—which also carries the retail leg the corridor never performs—is about 49%. A wholesale-plus-freight margin sits in the lower part of that range: the recorded c.i.f./f.o.b. freight and insurance gap on goods trade to the region is a single-digit percentage [Arvis et al., 2010], to which a wholesale distributive margin of a few points is added. We use **6–14%** and treat it as a calibration, not an estimate: because $\bar{v}^{TT} \approx \bar{v}^{M}$ the input–output step does no work, so the headline is a direct function of m and we carry it as a band. (We do not use the matched-cell unit-value wedge to pin m: censoring negative-margin cells inflates the gross margin to about 34%, while the value-weighted aggregate is negative, and both are artefacts of the c.i.f./f.o.b. basis mismatch and of outbound under-invoicing rather than measures of domestic value retained.) Domestic value added per dollar rerouted is then $m\,\bar{v}^{TT} \approx$ **5–11%**, midpoint about **8%**, against **76%** per dollar of domestic manufacturing output—so a rerouted dollar creates on the order of a tenth as much domestic value as a produced one. Of the $479m in incremental exports to Russia after 2022, Kazakhstan retains roughly **$23–53m** as domestic value added.

Because the propagation step is close to the identity, the headline is a one-for-one function of m, and the reader can substitute a different margin directly (Appendix D gives the full sweep, the input–output cross-check and the matched-cell wedge detail). Sweeping m across the full Bureau of National Statistics bracket: at the transport-only end ($m \approx 1\%$)

a rerouted dollar captures about one-hundredth of a produced one; across our 6–14% band, one-sixteenth to one-seventh; the "corridor, not factory" reading—a rerouted dollar worth on the order of a tenth of a produced one—holds for any m up to about 12%. It weakens to "about a fifth" only at $m \approx 19\%$, to "about a third" at $m \approx 32\%$, and reaches parity only at $m \approx 49\%$, the full trade-and-transport margin that includes the retail leg the corridor never performs. The censored matched-cell gross margin (34%), even taken at face value, still leaves a rerouted dollar at about a third of a produced one. The qualitative conclusion is thus robust to any margin a wholesale-plus-freight reading can support; it fails only if the corridor in fact captures a near-complete retail-inclusive markup. The one piece of evidence that bears directly on a markup that large is the value-weighted aggregate unit-value wedge, which is itself below one; we read that as suggestive rather than dispositive, because it is depressed by outbound under-invoicing and the c.i.f./f.o.b. basis mismatch.

The multipliers matter little by comparison. Even at $\bar{v}^M = 0.40$ the ratio is about one to five; recomputing $\bar{v}^{TT}$ and $\bar{v}^M$ from the Bureau of National Statistics symmetric input–output table (68 products, 2023) rather than the OECD ICIO gives $\bar{v}^{TT} = 0.89$ and $\bar{v}^M = 0.74$, moving the rerouted-vs-produced ratio from about one-in-ten to one-in-eight—still an order of magnitude. Under-invoicing moves it only so far: for domestic value added to reach one-fifth *of the gross rerouted flow*—a looser benchmark than one-fifth of a produced dollar, which $m \approx 19\%$ already breaches—the trade margin would have to be about 25%, which—holding the recorded import cost fixed—requires exports to Russia to be under-recorded by 13–21%. Under-recording on that scale is within the range found for transit trade [Fisman and Wei, 2004] and cannot be excluded; but even at that bound Kazakhstan retains about a quarter of what a producing economy captures, not more.

6.3 Fiscal and macro

The reorientation-attributable flow—incremental exports to Russia over the pre-2022 baseline—is about \$479m cumulatively over 2022–2025 (\$107–134m per year). Customs duty on it, at an EAEU common external tariff of 4–8% on the share formally cleared in Kazakhstan, is on the order of **\$22m over four years** (about \$5m per year); net import VAT is close to zero because the onward supply to Russia is intra-union and zero-rated. Kazakhstan's annual customs revenue is of the order of \$4–5bn, so the reorientation adds well under 1%. At the macro level the flow is 0.2–0.3% of GDP; the 2022 current-account surplus, the reserve accumulation of 2023–2025 and the rise in services value added are an order of magnitude too large to be driven by it.

7 The (non-)investment response

If the demand shock were seeding domestic value added, we would expect a rise in investment in the sectors the flow passes through. We find none—though, with only four post-2022 years, the deal-count test can rule out a large response, not a modest one.

Deal-making in manufacturing of tradeable goods, transport and logistics, and wholesale distribution runs at **7.6 per year** in 2015–2021 and **7.2 per year** in 2022–2025 (a Poisson rate ratio of 0.96, 95% CI 0.59–1.53); after 2015 and 2022—a coverage ramp-up year and the unrest year—2022 is the weakest year in the series (Figure 4). Excluding the thin 2015 first year the pre-2022 rate is 8.7 per year, so the post period is a slight decline on either basis. Dropping the 2022 trough, 2023–2025 runs at 8.7 per year (rate ratio 1.14, 95% CI 0.69–1.86). The test has 80% power only against an increase of roughly 80% or more, so it excludes a surge in deal-making but not a modest step-up. Three features of the data say more than the count. First, reported deal *value* in these sectors rises, but is concentrated in ownership transfers of existing or distressed assets—the ArcelorMittal Temirtau to Qarmet nationalisation (2023), Eurocopter Kazakhstan and the Lokomotiv rolling-stock plant (re-nationalisations, 2024), and a \$0.8bn 2025 consolidation of rail-freight and forwarding assets—not new productive capacity. Second, greenfield and new-capacity transactions average under two per year and never exceed four, with no post-2022 increase (one to two per year after 2022). Third, across the whole 2015–2025 window the data contain **no** transaction in the specific product lines that dominate the reorientation—though we do not lean on this, because the pre-2022 base rate in those narrow lines is also near zero, so the post-period zero carries little information on its own. Our adjudication rule is deliberately generous: we would have coded a positive investment response as any greenfield or capacity-expansion deal, in any of the three databases, tagged to a surge-basket HS6 *or* to its parent electrical-machinery, components or instrument-manufacturing class. On that rule the count is zero before and after; the informative statistic is the sector-level rate test above, which rules out a step-up of about a doubling or more.

The result holds across the three databases. Table 3 reports deal counts for the same Kazakhstan universe from each source; coverage differs in level—the transaction-oriented Capital IQ extract records more mid-market M&A, the venture-heavy PitchBook more small early-stage rounds, Preqin fewer of either—but none shows a post-2022 increase in manufacturing or logistics deal-making. In the Capital IQ extract, which best captures the mid-market industrial deals that a genuine capacity response would generate, the count of manufacturing, transport and distribution deals is essentially unchanged before and after 2022 (5.1 versus 5.2 per year).

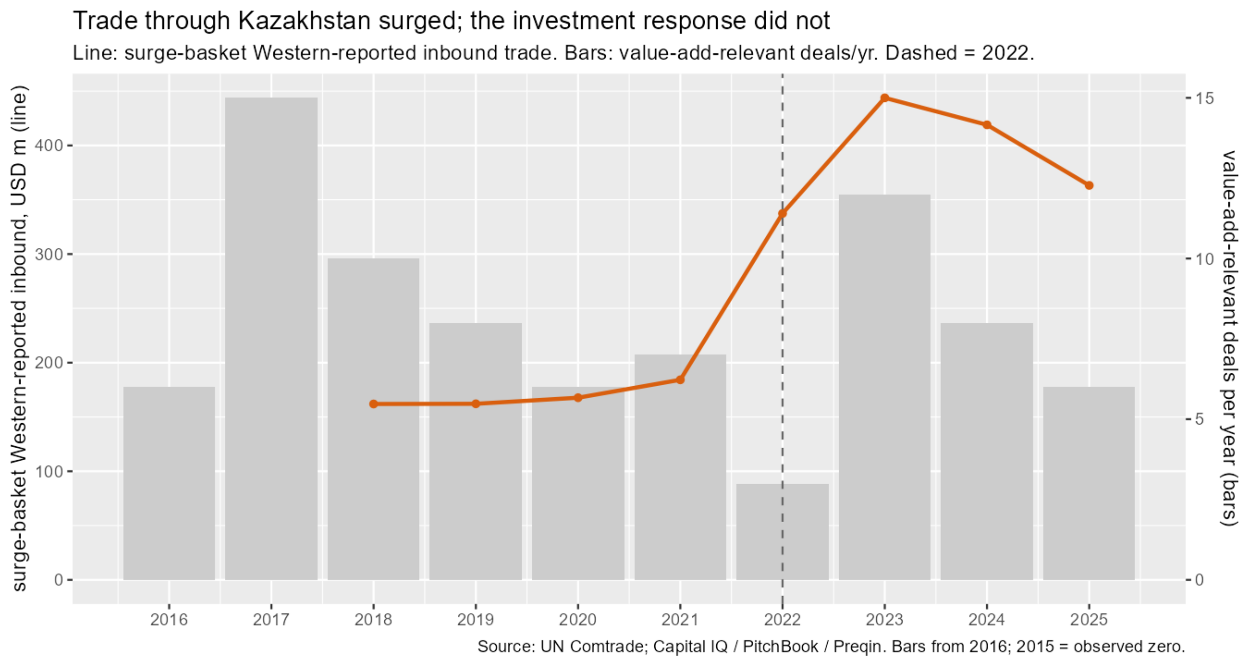

Figure 4: Trade through Kazakhstan surged; the investment response did not. Line: surge-basket *Western-reported* inbound trade (partner-reported mirror), USD million (left axis)—the West + China total is several times larger but was already growing before 2022. Bars: annual count of M&A/PE deals in manufacturing of tradeable goods, transport and logistics, and wholesale distribution (right axis). Dashed line: 2022. Bars shown from 2016; 2015 is an observed zero in the deal databases (Section 7). Source: UN Comtrade; Capital IQ/PitchBook/Preqin.

Kazakhstan is adding vehicle assembly (Chinese marques, KIA, Škoda) and some appliance capacity, but this predates and is largely independent of the reorientation, is kit assembly with low domestic value added, and serves the domestic and Eurasian consumer market rather than the re-export flow to Russia. The state investment corporation's disclosed post-2022 pipeline is domestic food, energy and infrastructure, not corridor processing [Qazaqstan Investment Corporation et al., 2026]. The one investment signal that is reorientation-linked—a 2025 consolidation of cross-border rail-freight and forwarding assets—deepens Kazakhstan's role as a forwarding operation, not a producing one.

8 Which gate binds?

The framework of Section 3 attributes the null to at least one closed gate. The market-access gate is plainly open: under the Eurasian Economic Union the good moves to Russia with no transformation and no border, so $V_T > 0$; and because a domestic plant's transformation margin sits below the thin re-export markup ($\mu_P \lesssim \mu_T$, on the sector's near-zero domestic base and its 0.69 value-added multiplier, Section 3), building is dominated in (1) before the

Table 3: Deal counts for the Kazakhstan universe, by source and period. Rates are per year over a 7-year pre-window (2015–21) and a 4-year post-window (2022–25). The consolidated non-duplicate universe is $N = 493$ transactions; the manufacturing, transport and distribution subset is $N = 82$ (53 pre, 29 post).

Deals per year (unless noted)	Cap. IQ	PitchBk	Preqin
All deals, 2015–21	24.0	14.6	3.7
All deals, 2022–25	23.2	24.2	1.8
Mfg + transport + distrib., 2015–21	5.1	2.0	0.4
Mfg + transport + distrib., 2022–25	5.2	1.5	0.5
Surge-basket lines, 2015–25 (total)	0[a]	0	0

[a] Two Capital IQ records carry an "electronic" industry tag—a steel-wire-rope manufacturer and a finance holding company—and are misclassifications; there are no transactions in the reorientation's product lines.

option value is added. The re-export flow is in any case not distinguishable in price or weight from ordinary Kazakh trade (Section 6). In a multiplicative decomposition that is close to sufficient for $R \approx 0$ on its own, and it limits what the case can say about the other two gates: we observe a single configuration, with no gate clearly favourable and the three not separately identified. What we can add is a state-fund pipeline that *rules out* the institutional gate as the binding constraint for this null (the least capital-constrained investor in the economy did not build either), and a durable-versus-uncertain shock comparison that is *suggestive* of an adverse irreversibility gate but confounded. Neither is a clean identification, and we set out the limits of each.

The state fund's disclosed pipeline bypasses the reorientation. Kazakhstan's state investment corporation (QIC/Baiterek) faces neither an external-finance nor an exit constraint, so the institutional gate does not bind for it the way it binds for an outside investor. Its deployment in fact *rose* sharply after 2022—about \$2.2bn cumulative over 2015–2025, some \$1.5bn of it in 2025, of which \$1.0bn was a single steel-producer bond [Qazaqstan Investment Corporation et al., 2026]. But the projects the report identifies are a power plant, a poultry producer, steel and energy bonds, a pipe-systems maker, a bioethanol plant, a school and an office block—domestic food, energy and infrastructure, none in a surge-basket line and none in corridor logistics built for the flow; the report's note that manufacturing became prominent in the deal record from 2023 refers to these domestic-market projects. This is illustrative rather than a controlled test—the report gives no pre-2022 baseline and no sector-level denominator—but the least capital-constrained investor in the economy did not build for the reorientation either.

A durable shock, the same institutions, a different outcome. Over the same window, Kazakhstan's *durable* passenger-vehicle demand shock—Western marques withdrawing from Russia, a domestic consumption boom, and Eurasian market access—was met with real capacity: a combined Changan/Haval/Chery assembly facility of about 90,000 units, a KIA plant (about \$200m), local assembly of several Škoda models, and a car-multimedia plant in Almaty (2024). The re-export shock was met with no plant at all. We treat this as illustrative rather than as a controlled test, for three reasons. First, in the deal databases auto-sector transactions actually *fell* after 2022 (six in 2015–2021, three in 2022–2025); the capacity evidence is from plant announcements and public reporting, not from the deal data, and no symmetric announcement search was run for the component sector—so this compares a searched sample against an unsearched one. Second, Section 7 notes that much of the auto capacity predates the reorientation. Third, the two shocks differ not only in persistence and certainty but in sunk-cost intensity, in whether market access requires domestic content (EAEU local-content rules bind for vehicles, not for the re-export—so the comparison also varies the market-access gate), and in the destination market. The comparison is consistent with a durable, transformation-requiring shock drawing capacity where a re-export shock does not; it is not a clean experiment on any single gate.

The re-export shock is uncertain, not clearly transitory. The flow to Russia peaked in September 2022 and fell by more than a third over the second half of 2023, from about \$15m per month in 2023-H1 to about \$9m in 2023-H2, as trade-monitoring measures tightened (the EU's eleventh sanctions package, Council Regulation (EU) 2023/1214 of 23 June 2023, introduced a transit-monitoring provision). But that within-year decline is a correction from the September-2022 spike to a plateau, not a decay path: on the *annual* series exports to Russia in these lines run \$128m, \$145m, \$119m and \$133m across 2022–2025—roughly ten times the pre-2022 baseline, with no downward trend and 2025 above 2024 (Table 1; Kazakhstan stopped monthly Comtrade reporting after February 2024, so the only monthly evidence covers a single half-year). *Ex post* the flow persisted. What a 2022 entrant faced was therefore high *uncertainty* rather than observed transience: the flow is policy-contingent (successive sanctions packages, monitoring provisions, compliance exposure) and its duration was unknowable. In the framework's terms this is a high-σ configuration, which raises Ω and delivers the same null as a low-ρ one; we do not claim the realised persistence was low.

The institutional gate binds on the adjacent opportunities. The state fund's pipeline rules the institutional gate out for the *shock-specific* null, but there is one place in the case where a single gate varies while the others hold, and it points to institutions: the durable adjacent opportunities of Section 10, for which the irreversibility gate is plausibly open (warehousing and transport support, with a domestic value-added multiplier of 0.82; machinery and fabricated-metal import substitution), are also financed only slowly and only by directed state capital. An open irreversibility gate with no private investment response is evidence that the institutional gate binds *there*. The domestic private-equity market has

Table 4: Moderators of the investment response, and Kazakhstan's values

Moderator	Gate	Kazakhstan, 2022	$R\uparrow$ when
Shock persistence ρ (structural reallocation vs. event-driven)	irreversibility	flow persisted *ex post*; *ex ante* uncertain	high
Uncertainty σ / policy risk	irreversibility	high (policy-contingent)	low
Sunk-cost intensity I of the relevant technology	irreversibility	high (component mfg)	low
Domestic transformation required to reach the market (rules of origin, local content, a border)	market access	none (customs union)	required
Financial depth / exit availability	institutions	shallow, no exit	deep
Private vs. state / directed capital mix	institutions	state-dominated	private-led
Existing productive base / absorptive capacity	institutions	near-zero (electronics output $\approx$ \$230m)	a base exists

no functioning exit, with the founder buy-back the leading route [Qazaqstan Investment Corporation et al., 2026]; capital is dominated by state and development-finance money, and the 2022–2025 pipeline is crowded by nationalisations. This is the institutional-voids configuration of Khanna and Palepu [2000] and Rajan and Zingales [1998], and it is the subject of related work.

9 Moderators and external validity

The decomposition organises seven moderators, listed in Table 4 with the value each takes for Kazakhstan. Its empirical content is in the three necessary conditions, not in the functional form of (2): R is positive only when the market-access and irreversibility gates are both open, and large only when the institutional gate is open as well. A country adverse on the first two should show a null irrespective of its institutions, and a country favourable on all three should show a supply response even with a shallow capital market.

Within the reorientation group. Armenia and the Kyrgyz Republic received the same shock—their surge-basket exports to Russia rose several-fold, breaking in 2022 on the same timing—and share Kazakhstan's value on every moderator: the shock is event-driven, its persistence uncertain *ex ante*, policy risk is high, and the customs union removes the need to transform. Two features are informative. First, both neighbours' flows had already receded by 2024–2025 (Armenia's surge-basket exports to Russia: \$71m, \$93m, \$42m, \$13m over 2022–2025; the Kyrgyz Republic's \$12m, \$30m, \$12m, \$12m), whereas Kazakhstan's persist (\$128m, \$145m, \$119m, \$133m). The neighbour reversion shows the flow could recede; the Kazakhstan asymmetry means the "same shock" claim holds for the 2022–2023 surge, not for its persistence. Second, we do not have deal-level data for Armenia or the Kyrgyz Republic—our commercial extracts are Kazakhstan-only—so we cannot test the investment null for them directly; what we can say is that neither has a domestic component-manufacturing sector to expand.

Outside the customs union. The named population of post-2022 intermediaries is {Kazakhstan, Armenia, the Kyrgyz Republic, Belarus} inside the Eurasian Economic Union and {Georgia, Türkiye} outside it. The distinction matters for the framework: a non-member routes goods to Russia across a customs border, so its market-access gate is not fully open, and the framework predicts it is the more likely of the two groups to see a domestic processing or assembly response. On the trade side the shock is visible in the non-members too—Türkiye's surge-basket exports to Russia roughly double at the 2022 break (from about \$58m in 2021 to \$119m in 2022 and \$207m in 2023; sup-F of 11.9, $p = 0.007$), while Georgia barely participates in this particular flow (under \$2m a year throughout, no clean break); Appendix E gives the full series and break tests, alongside Armenia and the Kyrgyz Republic. We cannot run the investment test for either: our deal data are Kazakhstan-only, and Türkiye's large pre-existing manufacturing base would in any case make a marginal-capacity response hard to detect in aggregates. The external-validity claim we can support is therefore narrow—the *demand shock* generalises across the intermediary population—while the *investment null* is established only for Kazakhstan, where the market-access gate is fully open and a domestic component sector barely exists.

Contrast cases. Economies that drew capacity investment from a post-2020 tradeable demand shock did so with a favourable configuration on the first two gates, not necessarily the third. Vietnam and Mexico received China-plus-one and nearshoring reallocations widely read as *structural* [e.g. Juhász et al., 2024], and in autos the tightening of rules of origin under the United States–Mexico–Canada Agreement made domestic content a condition of market access (a closed market-access gate that forces $V_T \to 0$). Both invested in capacity despite financial markets that are not uniformly deep, which is the pattern the framework predicts: persistence and a transformation requirement do the work, financial development scales the response rather than gating it.

What we cannot show here. A clean separation of the irreversibility and institutional gates across countries would regress sector-level greenfield investment on shock persistence and financial depth in a panel of intermediaries and contrast cases. In the sample available to us the two are collinear—Kazakhstan, Armenia, the Kyrgyz Republic and the other Caucasus and Central Asian intermediaries are all financially shallow (private credit plus market capitalisation of 24–66% of GDP, all below the comparator median); Türkiye is financially deeper but has too large a pre-existing manufacturing base for a marginal-capacity response to register—and country-level macro aggregates are too noisy to isolate a response of this size and in places run the other way: Armenia's gross fixed capital formation actually rose about three points of GDP after 2022. The within-country comparisons of Section 8 are what the argument leans on; a cross-country panel using greenfield-project data by sector is the natural extension.

10 Where investment would raise value capture

The reorientation is nonetheless informative for investment policy, because the trade data reveal which products have proven demand and proven routes through Kazakhstan. Table 5 sets three quantities side by side for the candidate sectors—the domestic value-added multiplier, the import-substitution gap (imports relative to domestic output), and the post-2022 growth in import demand—after removing the surge-basket lines from the electronics-adjacent chapters; we read them jointly rather than collapsing them into a single index (Figure 5).

Two groups stand out. The **logistics platform**—warehousing and transport support, and wholesale-trade infrastructure—has among the highest domestic value-added multipliers in the economy (0.82) and is the corridor's binding constraint, given the high overland logistics costs a landlocked transit economy faces [Arvis et al., 2010]; investment here creates domestic value and holds it whether or not the Russia flow persists. **Import substitution** in electrical equipment, machinery, fabricated metal products and chemicals: Kazakhstan imports six to eight times what it produces in electrical equipment and machinery and roughly twice its output in the other two; the domestic value-added multiplier is around 0.80, and a nascent deal base exists. By contrast, localising the surge-basket products is the worst option—computer, electronic and optical manufacturing has among the lowest value-added multipliers in the economy (0.69, tied with motor-vehicle kit assembly) and a near-zero domestic base.

These opportunities are investable at scale only if the state investment corporation shifts from lead direct investor to co-investor behind private leads, with larger syndicated tickets and a credible exit—trade sale or listing—rather than the buy-back that currently dominates. That is a question about capital-market institutions, taken up in related work.

Table 5: Sector characteristics for value-adding investment, Kazakhstan. VA mult. = domestic value-added multiplier $(v_d' L_d)_j$ from the OECD ICIO block; Output = gross output, \$m; Imp./output = 2023 imports over domestic output; Imp. growth = ratio of 2023 to 2019 import value. Dashes: not applicable (platform sectors are not import-competing) or not computed. Surge-basket HS6 are netted out of the electronics and machinery chapters. The substitution rows are the four top-ranked candidates on a combined score (value-added multiplier, import-substitution gap, import growth, deal base) out of $N = 13$ ranked tradeable sectors; two platform and two "avoid" rows are shown alongside, so 8 of the 13 appear here and the full ranking is in the replication output.

	Sector	VA mult.	Output \$m	Imp./ output	Imp. growth
Platform	Warehousing & transport support	0.82	6,800	—	—
	Wholesale & retail trade infrastr.	0.82	81,000	—	—
Substitution	Electrical equipment	0.81	1,150	5.6	1.43
	Machinery & equipment n.e.c.	0.80	1,160	7.9	1.27
	Fabricated metal products	0.81	1,850	1.8	1.22
	Chemicals	0.79	3,480	1.7	1.52
Avoid	Computer, electronic & optical (surge basket)	0.69	230	—	—
	Motor vehicles & parts (kit assembly)	0.69	1,530	6.1	2.55

11 Policy implications

The framework of Section 3 is also a menu of levers. A demand shock of this kind industrialises a transit economy only if at least one of three things holds: the market-access gate is closed to a pure re-export, the irreversibility gate is opened, or the institutional gate is opened. This section takes each in turn for Kazakhstan, then asks what its position on the China–Europe land route adds, and what its principal supplier could do. The discussion is normative and forward-looking; it is separate from the paper's positive results and rests on the framework rather than on new estimation.

11.1 Kazakhstan: raising the investment response

Market access. The customs union makes trade a near-perfect substitute for production, which is close to sufficient on its own for a null. The corresponding lever is to make some domestic value-added a condition of capturing the flow: procurement and investment-incentive rules that reward domestic processing, and use of the customs-union position to attract assembly rather than only transit. The target is not the surge-basket goods themselves—the

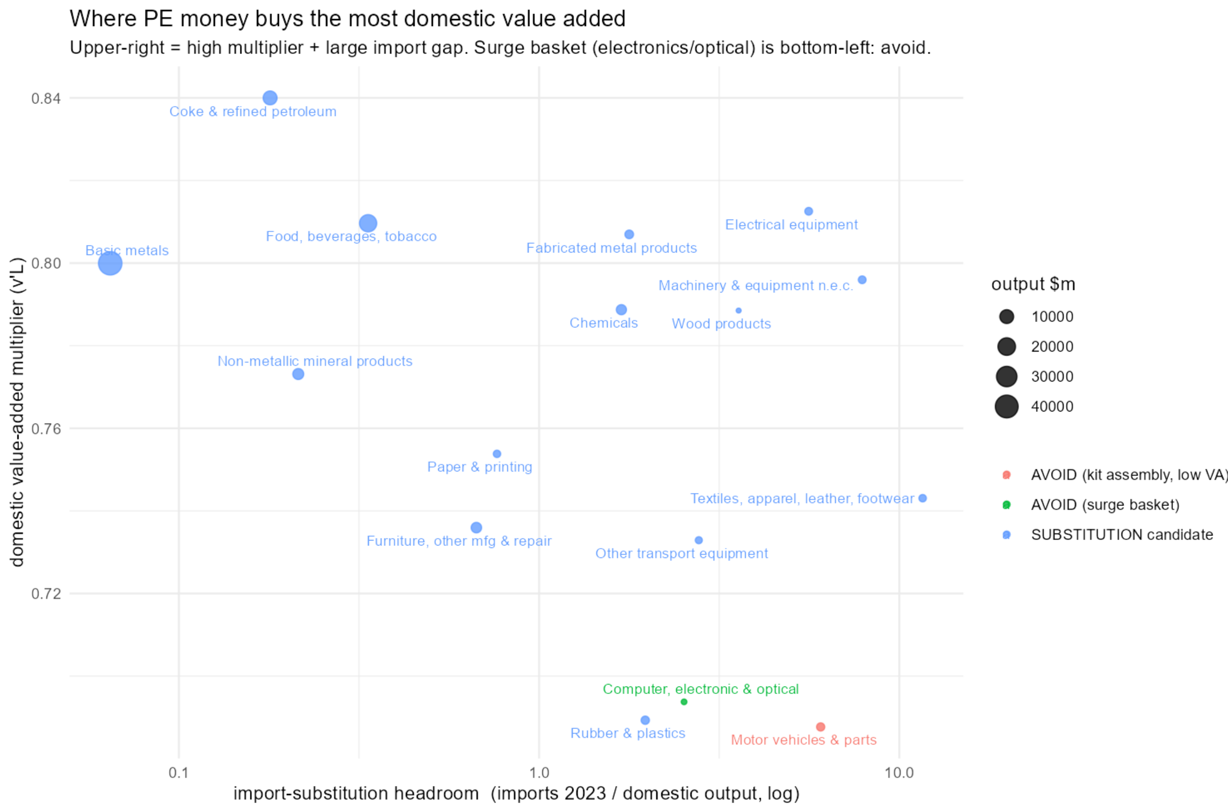

Figure 5: Where investment buys the most domestic value added: domestic value-added multiplier against import-substitution headroom (imports 2023 / domestic output, log scale), bubble size = domestic output. The surge-basket products (computer/electronic/optical) sit at the bottom.

"Avoid" row of Table 5, where domestic assembly of imported kits earns a low multiplier (0.69) and forcing it would mostly deter the throughput—but the substitution and platform sectors, and the *next* shock of this kind: a standing rule that flows above a threshold trigger a domestic-content review channels investment toward the 0.80-multiplier lines rather than toward low-value final assembly. This is the sharpest lever and the hardest, for two reasons. It trades some current throughput for the chance of durable capacity. And a unilateral local-content requirement on *onward* trade sits uneasily with the Eurasian Economic Union's common-market rules on the free movement of goods [Eurasian Economic Union, 2014], so the feasible form is a domestic incentive or procurement preference, or a measure agreed at union level, not a border restriction Kazakhstan can impose alone.

Irreversibility. Building beats waiting only when the shock is durable and certain enough and the technology cheap enough to sink. Policy can move two of these three. It can lower the effective uncertainty σ with multi-year commitments and credible demand signals—anchor and offtake contracts, state-backed demand aggregation across ministries and state firms, and a stable rule for how long an incentive lasts. It can lower the effective sunk cost I with shared pilot-line and testing infrastructure, ready-built special-economic-zone tenancy, and co-financing of first-of-kind facilities so that no single entrant carries the whole irreversible bet. The tension in this lever is fiscal risk-shifting: a state-backed offtake or demand-aggregation contract moves the demand risk from the investor to the budget, and if the flow stops the loss falls on the taxpayer rather than the firm. Given the 2022–2025 nationalisation wave, which already concentrated industrial risk on the state balance sheet, the defensible version is a capped, time-limited and partial guarantee tied to domestic value-added milestones, not an open-ended purchase commitment.

Institutions. The durable adjacent opportunities the shock reveals—logistics and transport support, machinery and fabricated-metal import substitution—are financed only slowly and only by directed state capital. The lever is to move the state investment corporation from lead direct investor to anchor limited partner and co-investor behind independent managers, with larger syndicated tickets and a credible exit route (trade sale or a public listing) rather than the founder buy-back that dominates today [Qazaqstan Investment Corporation et al., 2026], and to build the domestic private-equity exit market alongside it.

The platform first. Across the priority table the highest-return use of investment is the logistics platform itself—warehousing, transport support and Trans-Caspian capacity—which has among the highest domestic value-added multipliers in the economy (≈ 0.82) and creates value that holds whether or not any particular flow persists.

11.2 Corridor leverage in the current geography

Kazakhstan's bargaining position on these questions is stronger than the value-capture numbers alone suggest. Since 2022 the northern land corridor through Russia has been closed to most Western supply chains, and over 2023–2025 the principal maritime routes were disrupted in turn—draft restrictions on the Panama Canal, re-routing around the Red Sea, and periodic risk in the Strait of Hormuz. Each raises the value of an overland China–Europe alternative, and the shortest route that avoids both Russia and the maritime chokepoints runs through Kazakhstan and across the Caspian. Kazakhstan is close to an unavoidable node on that route, which is a structural bargaining chip with China, which wants route diversification and resilience, and with the European Union, which wants a land option that does not run through Russia.

The long-run play, in the framework's terms, is to use corridor access, capacity and reliability guarantees as the lever the customs union otherwise removes: to bundle throughput and priority-handling commitments with requirements or incentives for co-located processing, local content, joint-venture assembly, and standards and technology transfer—converting transit rent into value-added capacity rather than collecting fees for pure pass-through. It is the same recommendation as the market-access lever above, reached from the geography rather than from the customs-union rule.

The rent itself is small; the leverage is in the option, not the fee. The Trans-Caspian route carried on the order of 2–3 million tonnes in 2023–2024 and is targeted at roughly 10 million by 2030 [World Bank, 2023], against China–EU merchandise trade of some $800bn a year; transit and handling charges on even the 2030 volume are a fraction of one percent of Kazakh GDP. What makes the route a bargaining chip is that it is the marginal land link China and the EU would need if the maritime disruptions persisted, not the revenue it currently produces.

Three caveats bound this. Transit demand is contestable: a southern route through Iran and Türkiye and competing Caspian nodes in Azerbaijan and Georgia would capture part of any rent. Caspian ferry and port throughput and the rail break-of-gauge are binding physical constraints on how much traffic the route can actually take. And the leverage lasts only as long as the chokepoint disruptions and China's route diversification remain incomplete.

11.3 China: lowering barriers to Kazakhstan capturing more value

Because China supplies about two-thirds of the inbound flow, some of the levers sit on the Chinese side. These are hypotheses the paper cannot test, offered as directions rather than findings. Value-added-tax rebate and export-processing rules—a major instrument of Chinese industrial policy, with rebate rates that vary sharply by product and demonstrably shift export composition [Gourdon et al., 2022]—currently favour finished-goods export over component export plus assembly in Kazakhstan, tilting firms toward shipping the completed article rather than locating a stage of production on the route. Approval friction for mid-sized outward-investment joint ventures in Central Asia raises the fixed cost of the kind of plant that would add domestic value. Mutual recognition of standards and certification would let a Kazakhstan-assembled good qualify for both the Chinese and the Eurasian market without a second conformity process. And coordination of bonded-processing zones under the Trans-Caspian and Belt-and-Road frameworks would give co-located assembly the same customs treatment as pure transit.

None of these levers substitutes for the others: the market-access and corridor-leverage levers are the most powerful and the hardest to pull, and the institutional lever is a precondition for private capital to act once they are.

12 Discussion

Kazakhstan's trade with Russia and with the West in these products moved by an order of magnitude after 2022. The domestic economic content of that movement—value added about a tenth of the manufacturing equivalent, customs duty under one percent of revenue, a fraction of one percent of GDP, and no detectable investment response—is small. The country is functioning as a corridor, not a site of production.

The result speaks to a general question: when does a trade shock industrialise a country, and when does it merely pass through? The framework of Section 3 gives a three-part answer. Production is a plausible response only when the demand cannot be met by trade alone (a closed market-access gate) and only when the shock is durable and certain enough, and the technology cheap enough to sink, that building beats waiting (an open irreversibility gate); and it is actually realised only where the capital market can fund the projects that pass (an open institutional gate). What the Kazakhstan case identifies is the first condition and one half of the third: the customs union makes trade a close substitute for production, so the shock is absorbed into the trade channel; and the state investment corporation, which does not face the capital-market constraint, did not build either, which rules the institutional gate out as the binding constraint for this null (though its mandate and its cross-border-compliance exposure cloud the comparison). Whether the irreversibility gate *independently* binds the case cannot say—the re-export flow persisted through 2025, so what deterred a 2022 entrant was uncertainty rather than observed transience, and the durable-versus-uncertain shock comparison varies more than one gate at a time. The institutional gate does explain the missed alternative: the durable adjacent opportunities the shock reveals, in logistics and machinery, are left to slow, directed state capital, and go unfunded by private capital even though their irreversibility gate is open. Armenia and the Kyrgyz Republic show the same 2022 surge and, on the evidence we have, no domestic component-manufacturing sector to expand; the framework's contrast cases—structural reallocations, or rules-of-origin regimes that force domestic content—are where a supply response does occur, even without a deep capital market. The static value Kazakhstan misses—value added a tenth of the manufacturing equivalent—is the smaller part of the cost. The larger part is dynamic. A rules-of-origin or local-content regime that forced a stage of production onshore would also create the backward linkages through which foreign suppliers transfer technology and raise the productivity of domestic firms [Javorcik, 2004], and the supplier learning and diffusion that come with a durable production relationship [de Souza et al., 2024]. A pure corridor forgoes all of this: the goods pass through without a domestic firm ever entering the supply chain, so there is no linkage to learn from. That is where the long-run cost of being a corridor sits, and it is not captured by the value-added arithmetic.

A contemporaneous manufacturing expansion—and why it is not a counterexample. Over the same window Kazakhstan's aggregate manufacturing sector did grow: for January–July 2026 it accounted for about 47% of industrial output against 46% for mining—the second year manufacturing has run ahead—with manufacturing up about 9% year on year while mining fell about 4% and crude-oil output about 9% [Bureau of National Statistics, Agency for Strategic Planning and Reforms of the Republic of Kazakhstan, 2026]. Read against this paper the expansion is consistent with the framework rather than against it, for three reasons. First, part of the crossover is the denominator: mining and oil contracted, so a rising manufacturing *share* overstates the real divergence. Second, the growth is concentrated in passenger vehicles and assembly (machinery output up about 22%, on full-cycle plants with roughly 190,000 units of announced capacity) and in domestic-demand import substitution (fabricated metals, chemicals, pharmaceuticals)—not in the re-export lines of this study, where no capacity was added. Third, the mechanism is the one the framework names: minimum local-content shares written into subsoil-user contracts, a shift to mandatory procurement from domestic producers by large firms and state agencies, off-take contracts underwriting demand, and directed state and development finance behind new plants [Government of the Republic of Kazakhstan, 2026, Eurasian Economic Union, 2014]. That is a deliberately *closed* market-access gate plus directed institutions—exactly the configuration under which the model predicts a supply response. Where Kazakhstan required domestic content or supplied the demand itself, investment followed; where the good could move onward untransformed inside the customs union, it did not. The contrast is the paper's point, not a challenge to it.

Limitations. Four limitations are central. First, the case observes a single configuration on which the three gates are not separately identified, so it identifies the market-access gate as open and rules the capital-market institutional gate out as the binding constraint for the shock-specific null, but it cannot establish that irreversibility independently binds; the durable-versus-uncertain shock comparison and the state-fund pipeline are suggestive corroboration, not clean experiments, and the cross-country panel that would separate the irreversibility and institutional gates requires greenfield-project data by sector and country. Second, the value-capture headline is a calibration: because the input–output step does almost no work, it scales one-for-one with a trade margin we choose rather than estimate (Section 6); the "corridor, not factory" reading—a rerouted dollar worth about a tenth of a produced one—holds for a margin up to about an eighth, weakening to "about a fifth" at a margin near a fifth. Third, the investment null is *over*-determined—the January 2022 unrest, the cross-border compliance exposure noted in Section 2, and the nationalisation wave would each depress entry—so we bound its magnitude (Section 7) rather than attribute it to the trade shock alone. Fourth, the deal databases capture *transacted* investment—M&A and private financing rounds—and under-cover pure greenfield capacity that is built without an external transaction. A dedicated greenfield register (fDi Markets, or Orbis Cross-border Investment) would close that gap; the plant-level evidence we use in Section 8 is a manual substitute, and it too shows no reorientation-linked capacity.

Five measurement caveats bear on the estimates. The inbound side is partner-reported (mirror) data, which overstates Kazakh absorption for onward-moving goods. The inbound flow is dominated by a large pre-existing China component, so we measure value capture on the incremental Western flow and read the West + China inbound only at the product level, where line and year fixed effects absorb the China trend. The surge basket is selected on the post-2022 outcome; the externally defined priority list gives the same picture, but the cross-sectional difference-in-differences should be read alongside the structural breaks and the neighbour comparison rather than on its own, and a placebo on the largest civilian lines is significant with the opposite sign, indicating the control pool is not homogeneous in trend. The retained margin is taken from the national-accounts convention (6–14%), not estimated line by line, and the value-added multipliers come from the OECD ICIO (2019), cross-checked against the Kazakhstan national input–output table (2023) with the same qualitative result.

A Deal-database query specification

For online publication.

The Kazakhstan deal universe is extracted from three commercial databases (Capital IQ, PitchBook, Preqin). Common parameters: *target/issuer geography* = Kazakhstan (country of incorporation or primary operations); *announced date* between 1 January 2015 and 31 December 2025; *deal types* = merger/acquisition, majority/minority stake, private-equity buyout and growth, venture (all stages), joint venture, and PIPE/private placement; *status* = completed or announced (pending and withdrawn recorded separately). Financials in USD at announcement.

Capital IQ (S&P): Screening > Transactions; Target Company Location = Kazakhstan; Transaction Types as above; Announced Date range as above. Export all columns; extraction date recorded in the replication file.

PitchBook: Advanced Search > Deals; Company HQ Location = Kazakhstan; Deal Types = M&A, Buyout/LBO, PE Growth/Expansion, all VC stages, Angel, Accelerator/Incubator; Deal Date range as above.

Preqin: Private Equity > Deals & Exits; Portfolio Company Country = Kazakhstan; Deal Date range as above.

De-duplication is on target name, announced date (±30 days) and deal type. Sector classification is harmonised to ISIC rev. 4 divisions from each database's native taxonomy via a documented crosswalk. The three extracts, the crosswalk, and the reconciliation script are in the replication package, which pins each source's extraction date and its earliest year of reasonably complete Kazakhstan coverage.

B Product-set construction

For online publication.

The candidate set is 75 HS6 lines: 50 from an externally compiled list of technology-intensive items published in February 2024 (the "priority list"), grouped by its publisher into six tiers, plus 25 clearly civilian control lines. The **surge basket** is defined from the data as the lines in which the mean inbound flow (Western reporters + China) and the mean outflow to Russia *each* at least doubled from 2019–2021 to 2022–2024, with the ratio computed after adding $10,000 to numerator and denominator, and with post-period floors of $0.2m (inbound) and $0.1m (outbound). Twenty-nine lines qualify; 24 are on the priority list and 5 are control lines that also surged. Of the 75 candidates, 55 clear the outbound doubling and 41 the inbound one; requiring both is the binding screen.

C Difference-in-differences robustness

For online publication.

Table C.1 collects the full battery summarised in Section 5.3. The outcome throughout is asinh of Kazakhstan's exports to Russia; γ is the coefficient on treated $\times$ post with HS6 and year fixed effects and cluster-robust standard errors by HS6, unless noted.

D Value capture: input–output detail

For online publication.

Table B.1: Priority-list HS6 codes by publisher tier, and the civilian control set. † marks the 29 surge-basket lines.

Tier	HS6 codes
1	854231†, 854232, 854233†, 854239†
2	851762†, 852691†, 853221, 853224, 854800†
3A	847150, 850440, 851769, 852589†, 852910, 852990, 853669†, 853690, 854110†, 854121, 854129†, 854130, 854149†, 854151, 854159†, 854160
3B	848210, 848220†, 848230†, 848250, 880730, 901310†, 901380, 901420, 901480
4A	847180†, 848610, 848620, 848640, 853400, 854320†, 902750†, 903020, 903032†, 903039†, 903082
4B	845710†, 845811†, 845891†, 845961, 846693†
Control	090111, 170490, 190590, 210390, 210690, 220300, 330300, 330499, 340111, 392321, 392490†, 401110†, 480256, 610910†, 611020†, 620342, 630260, 640399, 691110, 732690, 841810, 870899, 940360, 950300, 961900†

Priority list: EU/US/UK/JP List of Common High Priority Items, version of February 2024. Tier labels are the publisher's; we carry them unchanged.

Table C.1: Difference-in-differences robustness, surge basket, exports to Russia.

Specification	γ	s.e.	p
Baseline (HS6 + year FE)	2.44	0.96	0.013
wild-cluster bootstrap p (B = 1999)			0.010
Holm-adjusted across 4 surge-basket outcomes			0.039
+ size-decile × year FE	2.29		0.003
Drop 2022 entirely	2.71	1.05	0.012
Donut (drop 2022, treatment year = 2023)	2.71	1.05	0.012
Poisson (PPML) in levels, e^{β}	3.5×		0.020
Priority list (50 HS6, externally defined)	2.88	0.74	< 0.001
Placebo (largest civilian lines)	−0.95	0.25	0.001
Leave-one-HS2-out jackknife			
Drop HS 39 (1 line)	2.51	0.98	0.013
Drop HS 40 (1 line)	2.48	0.98	0.014
Drop HS 61 (2 lines)	2.67	1.00	0.009
Drop HS 84 (7 lines)	2.28	1.18	0.058
Drop HS 85 (13 lines)	1.36	1.06	0.207
Drop HS 90 (4 lines)	3.16	0.96	0.002
Drop HS 96 (1 line)	2.59	0.97	0.010
Alternative surge-selection thresholds (both legs)			
1.5× (40 lines)	1.59	0.90	0.080
2.0× (29 lines, baseline)	2.44	0.96	0.013
2.5× (23 lines)	3.12	1.08	0.005
3.0× (18 lines)	3.16	1.27	0.015
Selection-rule-matched randomisation inference			
Free permutation (year labels within HS6, rule re-applied): the rule selects a mean of 5.0 lines under H_0 (max 12 over 1,000 draws), so $P(\text{rule selects} \geq 29 \mid H_0) = 0.000$; the permuted-null γ has mean 2.79, s.d. 1.50, 95th percentile 5.37, and $P(\lvert\gamma^{\text{perm}}\rvert \geq 2.44) = 0.58$. Trend-preserving cyclic-shift null: mean 5.2 lines selected (max 19), $P(\geq 29 \mid H_0) = 0.000$.			

Domestic value added per rerouted dollar is $m\,\bar{v}^{TT}$, against $\bar{v}^{M}$ per dollar of domestic manufacturing output. From the OECD ICIO Kazakhstan block, $\bar{v}^{TT} = 0.787$ (trade, transport, warehousing) and $\bar{v}^{M} = 0.764$ (manufacturing), so the rerouted-vs-produced ratio is $\approx 1.03\,m$ and the input–output step does almost no work. Recomputing on the Kazakhstan Bureau of National Statistics symmetric input–output table (2023, 57 industries matched, output-weighted within group) gives $\bar{v}^{TT} = 0.885$ and $\bar{v}^{M} = 0.742$, moving the ratio from 0.10 to 0.12. The BNS resources table brackets m: the transport-only margin on machinery and electronics is about 1% of basic-price value and the full trade-and-transport margin (which carries the retail leg the corridor never performs) about 49%; the 6–14% band is a freight-plus-wholesale reading, above the floor and well below the ceiling.

Table D.1: Sensitivity of the value-capture headline to the retained margin m. Ratio = domestic value added per rerouted dollar, relative to per produced dollar ($m\,\bar{v}^{TT}/\bar{v}^{M}$, OECD ICIO).

m	0.06	0.10	0.14	0.19
VA per rerouted \$	0.047	0.079	0.110	0.150
Ratio to a produced \$	0.062	0.103	0.144	0.196
Retained, \$m (of \$479m)	23	38	53	72
m	0.25	0.32	0.34	0.49
VA per rerouted \$	0.197	0.252	0.268	0.386
Ratio to a produced \$	0.258	0.330	0.350	0.505
Retained, \$m (of \$479m)	94	121	128	185

Anchors: transport-only $m \approx 0.01$; freight-plus-wholesale band 0.06–0.14; censored matched-cell gross margin 0.34; full trade-and-transport margin 0.49. The "corridor, not factory" reading (a rerouted dollar ≈ a tenth of a produced one) holds for m up to about 0.12; it weakens to "about a fifth" at $m \approx 0.19$, "about a third" at $m \approx 0.32$, and reaches parity at $m \approx 0.49$.

The matched-cell unit-value evidence (annual, $n = 112$ HS6×period cells) is descriptive only. The median re-export/import ratio is 0.73 on a c.i.f./f.o.b. basis and 1.6 on a like-for-like f.o.b. basis; the civilian control basket gives 0.59 and 0.94. The pass-through slope $\log(\mathrm{uv}_{\mathrm{expRU}})$ on $\log(\mathrm{uv}_{\mathrm{mirWC}})$ is 0.48 ($p = 0.03$), and 0.20 ($p = 0.46$) against the Kazakh-reported import price. The f.o.b. wedge by tier is 0.47 (Tier 1), 3.83 (2), 1.82 (3A), 1.09 (3B), 3.94 (4A), 1.35 (4B), against 1.33 for the control basket—high multiples concentrated in the most sensitive tiers, the pattern of a scarcity markup rather than a processing margin. The cell-level weight ratio (re-exported over imported tonnage) has median 0.08 and aggregate 0.11; the aggregate weight ratio equals the aggregate value flow-through (0.11), so a sub-one weight ratio is largely a flow-through fact. Eleven per cent of cells exceed 1.05, on an interquartile range of 0.02 to 0.49. In the six near-pure-transit cells (value flow-through in $[0.8, 1.25]$) the weight ratio is 0.74 at the median (0.26 and 0.94 at the quartiles).

E Neighbour and comparator structural breaks

For online publication.

Table E.1 reports surge-basket exports to Russia for the other post-2022 intermediaries and a mean-shift (sup-F) break test on the annual asinh series, 2018–2025. Armenia, the Kyrgyz Republic and Belarus are inside the Eurasian Economic Union with Russia; Georgia and Türkiye route across a customs border. The 2022 break appears regardless of customs-union status; the demand shock generalises across the population, while the investment test is Kazakhstan-only for want of deal data elsewhere.

Table E.1: Surge-basket exports to Russia, USD million per year, and the annual break test.

	2018	2019	2020	2021	2022	2023	2024	2025	sup-F	p
Kazakhstan (EAEU)	17	12	7	8	128	145	119	133	150.7	< 0.001
Armenia (EAEU)	15	9	7	9	71	93	42	13	10.5	0.015
Kyrgyz Republic (EAEU)	1	3	6	7	12	30	12	12	13.4	0.004
Türkiye (non-EAEU)	53	55	47	58	119	207	98	73	11.9	0.007
Georgia (non-EAEU)	0.2	0.7	0.4	0.7	1.0	1.1	1.7	0.7	6.9	0.075

Kazakhstan row from Table 1. sup-F is a mean-shift test on a persistent series; we read it as evidence of a break, not a cardinal magnitude.

Declaration of competing interest

The author is employed by the Astana International Financial Centre (AIFC), which operates the Astana International Exchange and is institutionally adjacent to Kazakhstan's state investment vehicles, and previously worked with a consulting team on a Kazakhstan trade-strategy engagement. The state-fund investment figures used here are drawn from a report co-published by the AIFC with the Qazaqstan Investment Corporation (QIC) and the International Finance Corporation (IFC). Neither role enters the empirical analysis, which rests on public data and on commercial deal databases available under academic licence; the policy recommendations in Section 11 are the author's own and do not represent the AIFC, QIC or the IFC.

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Data availability

The trade analysis uses public data: UN Comtrade (HS6 annual and monthly), World Bank World Development Indicators, the OECD Inter-Country Input–Output tables (2023 edition), and the Kazakhstan Bureau of National Statistics symmetric input–output and resources tables. The deal-level data are licensed from Capital IQ, PitchBook and Preqin and cannot be redistributed, but the full query specification (Appendix A) and the native source deal identifiers are in the replication package, so the extract is reconstructible by a licence holder. State-fund investment aggregates are from the published QIC/AIFC/IFC report. Code and query specifications are in the replication package, under `scripts/R/kz_passthrough/` and `scripts/R/kz_valueadd/`.

Declaration of generative AI in the writing process

During the preparation of this work the author used Claude (Anthropic) to draft and edit prose, to help write and debug the analysis code, and for structured internal review of the manuscript. All data choices, analytical decisions, results and interpretations are the author's own; the author reviewed and edited all output and takes full responsibility for the content of the publication.

CRediT authorship contribution statement

Zhanbolat Kakishev: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing — original draft, Writing — review & editing, Visualization.

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